

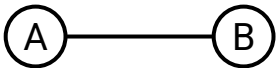
Bandwidth and Propagation Delay

Lecture 3, Spring 2026

Links

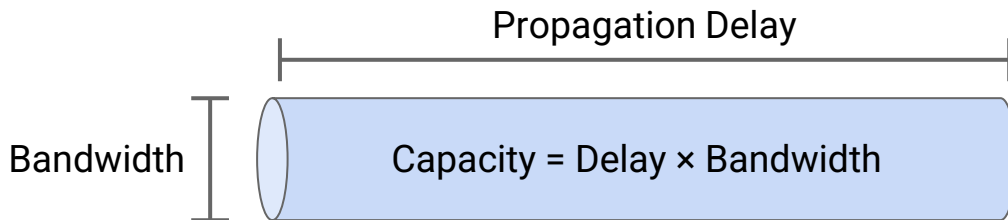
- **Bandwidth and Propagation Delay**
- Pipe Diagrams
- Overloaded Links

Properties of Links

A link connects two devices. 

Properties of a link:

- **Bandwidth:** Number of bits sent/received per unit time.
 - "Width" of the link.
 - Measured in bits per second (bps).
- **Propagation delay:** Time it takes a bit to travel along the link.
 - "Length" of the link.
 - Measured in seconds.
- **Bandwidth-delay product:** Bandwidth $\times$ delay.
 - "Capacity" of the link.



Measuring Packet Delay with Timing Diagrams

Suppose we have a link with:

- Bandwidth = 1 Mbps. (*1,000,000 bits per second.*)
- Propagation delay = 1 ms. (*0.001 seconds.*)

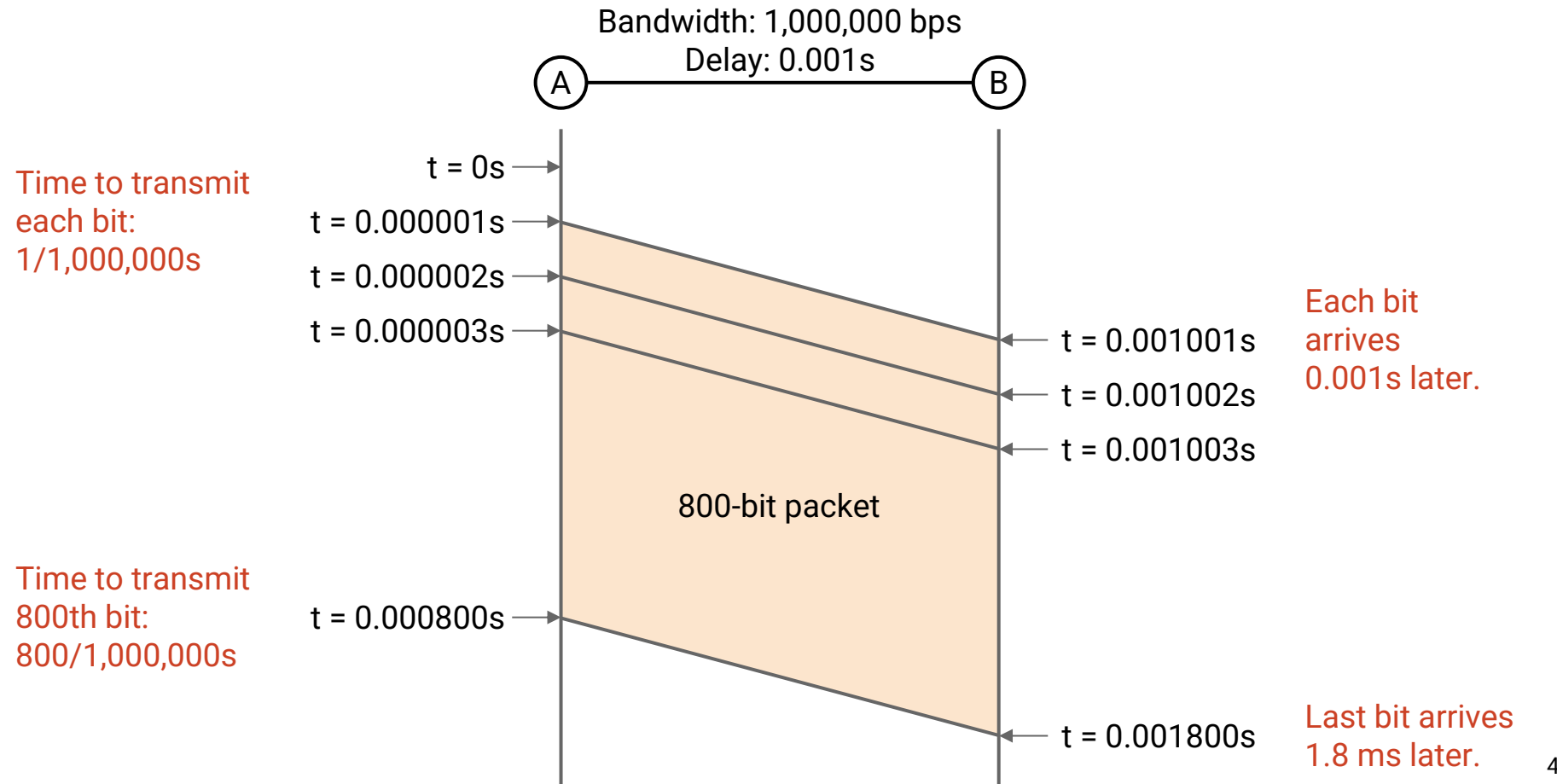
Note: We measure in **bits** per second, not **bytes**!

How long does it take to send a 100-byte (*800-bit*) packet?

- From the time the first bit is sent,
- To the time the last bit is received.

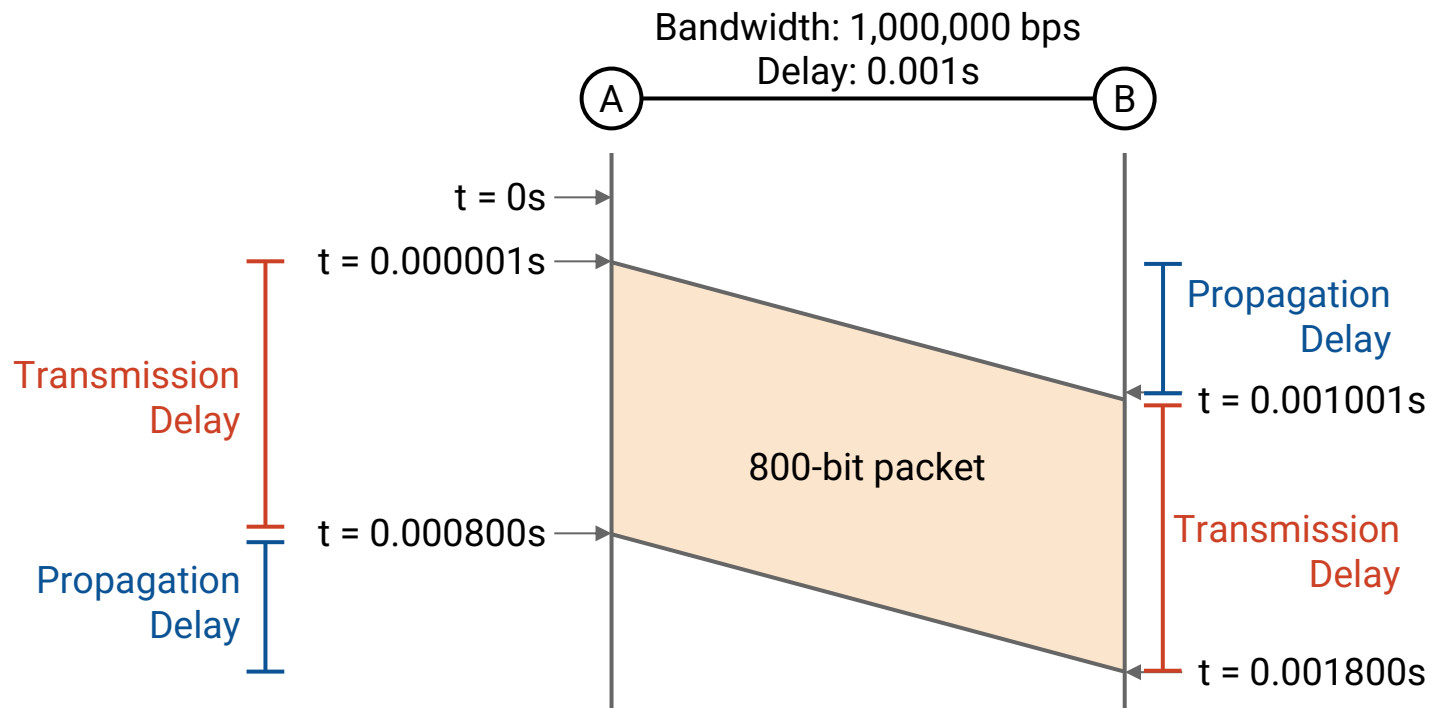
Let's draw a timing diagram to help.

Measuring Packet Delay with Timing Diagrams



Measuring Packet Delay with Timing Diagrams

$$\begin{aligned}\text{Packet Delay} &= \text{Transmission Delay} + \text{Propagation Delay} \\ &= (\text{Packet Size} / \text{Bandwidth}) + \text{Propagation Delay}\end{aligned}$$



Link Tradeoffs

Which link is better? It depends.

- Link 1: Bandwidth 10 Mbps Propagation Delay = 10 ms
- Link 2: Bandwidth 1 Mbps Propagation Delay = 1 ms

10-byte packet: Link 2 is better.

- ~10 ms with Link 1. ~1 ms with Link 2.
 - Link 1: $T_{tx} = 10 \text{ byte} / 10 \text{ Mbps} = 80 \text{ bits} / 10 \text{ Mbps} = 0.008 \text{ ms}$, $T_{prop} = 10 \text{ ms}$, Total delay = 10.008 ms
 - Link 2: $T_{tx} = 10 \text{ byte} / 1 \text{ Mbps} = 80 \text{ bits} / 1 \text{ Mbps} = 0.08 \text{ ms}$, $T_{prop} = 1 \text{ ms}$, Total delay = 1.08 ms
- For small packet, transmission delay is negligible. Propagation delay dominates.

10,000-byte packet: Link 1 is better.

- ~18 ms with Link 1. ~81 ms with Link 2.
 - Link 1: $T_{tx} = 10000 \text{ byte} / 10 \text{ Mbps} = 80000 \text{ bits} / 10 \text{ Mbps} = 8 \text{ ms}$, $T_{prop} = 10 \text{ ms}$, Total delay = 18 ms
 - Link 2: $T_{tx} = 10000 \text{ byte} / 1 \text{ Mbps} = 80000 \text{ bits} / 1 \text{ Mbps} = 80 \text{ ms}$, $T_{prop} = 1 \text{ ms}$, Total delay = 81 ms
- For large packet, transmission delay dominates.

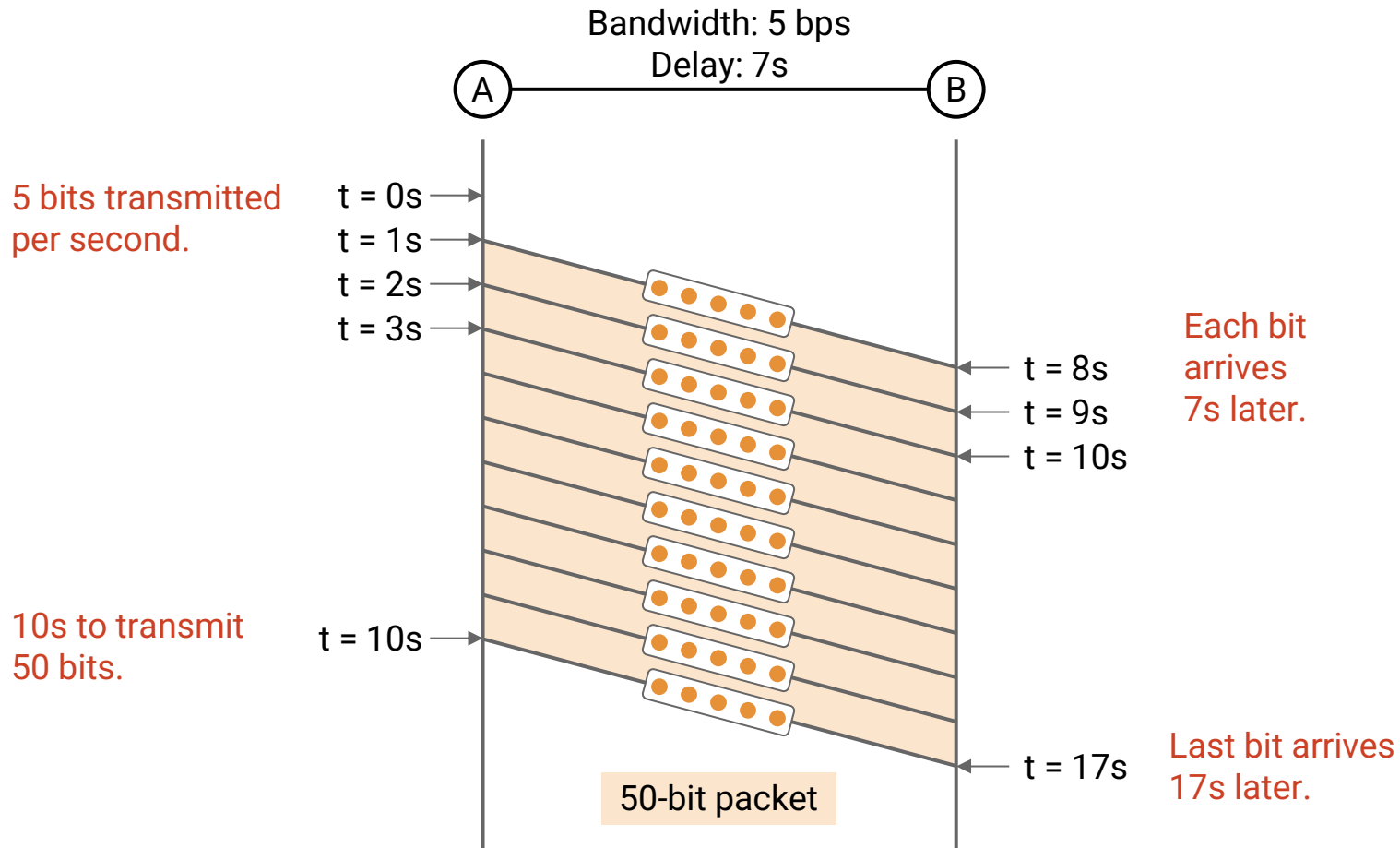
Pipe Diagrams

Lecture 3, Spring 2026

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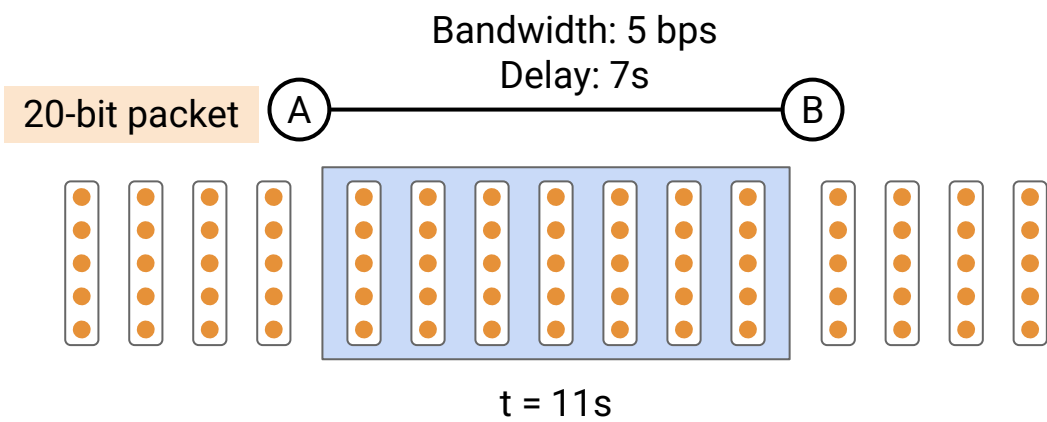
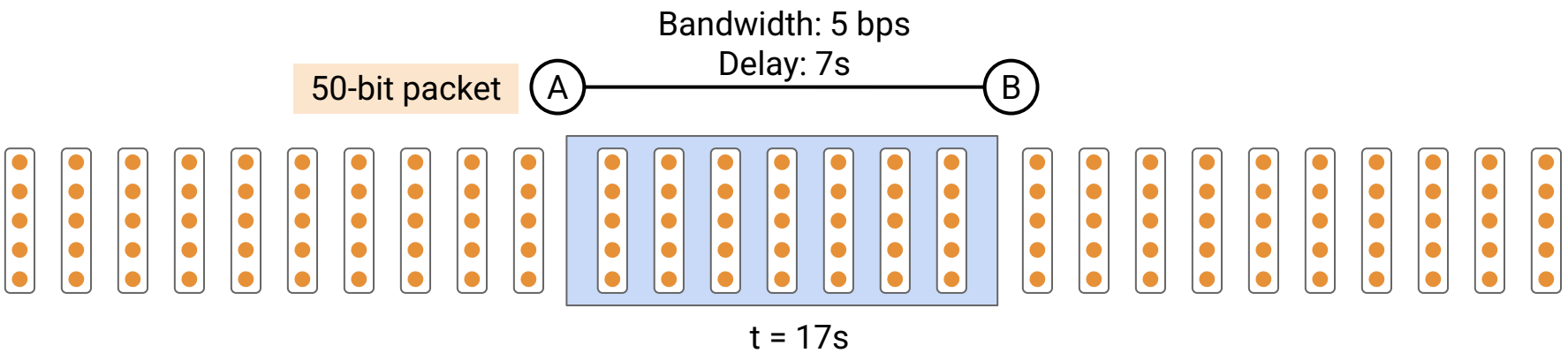
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Timing Diagrams and Pipe Diagrams



Timing Diagrams and Pipe Diagrams

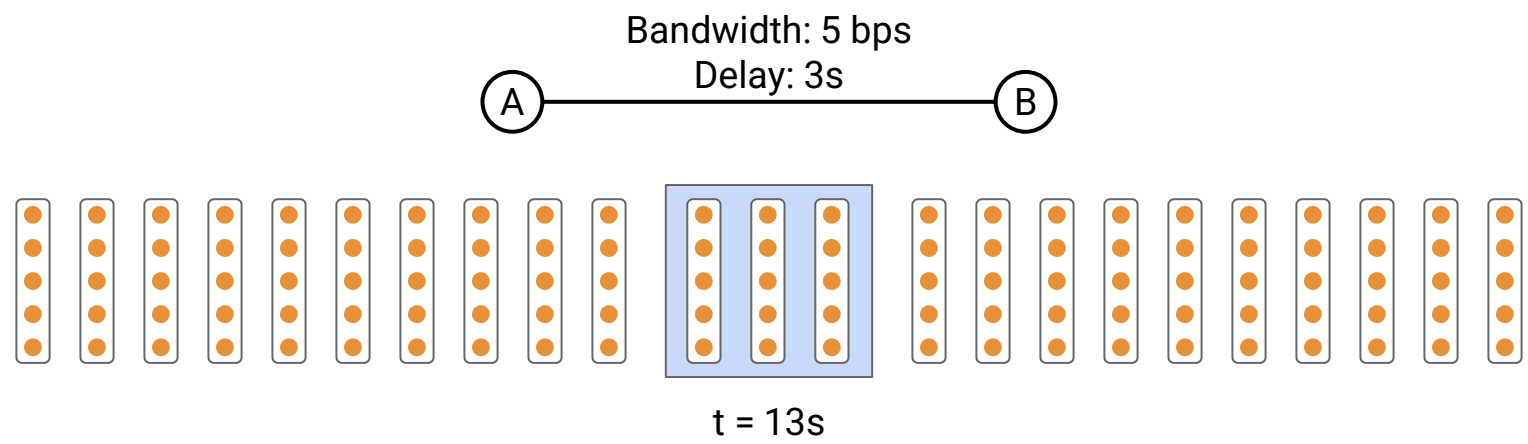
The pipe diagram shows the bits on the link at any given moment in time.



(These diagrams needsto be viewed with animation to make sense.)

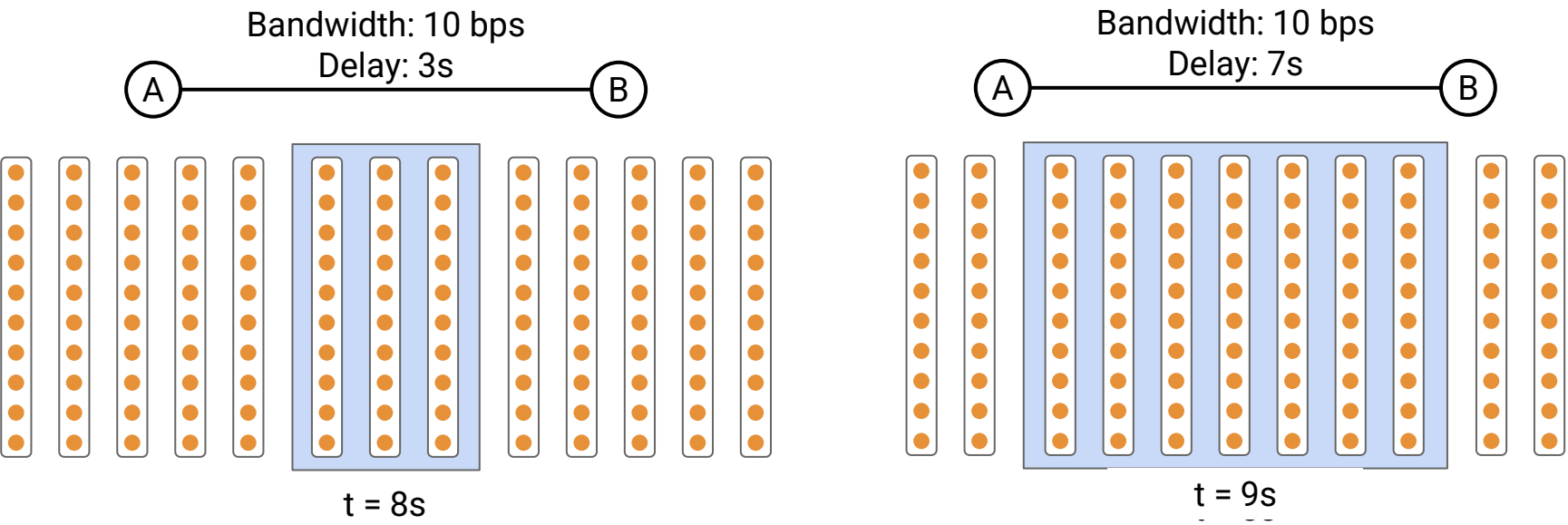
Pipe Diagrams

Shorter propagation delay: Pipe length is shorter.



(This diagram needs to be viewed with animation to make sense.)

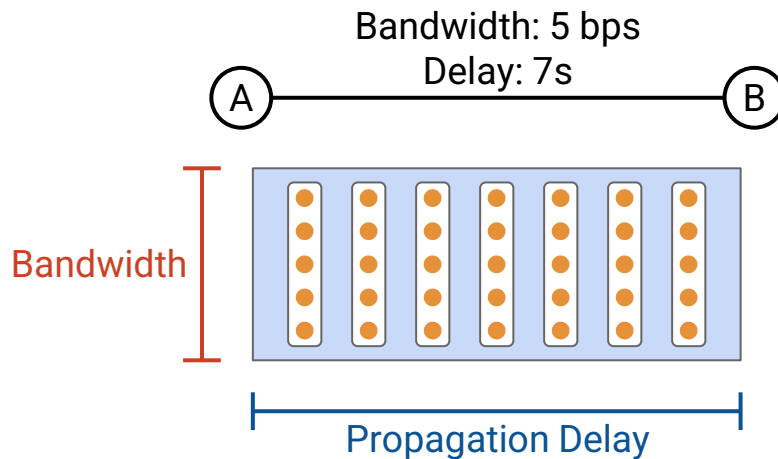
Higher bandwidth: Pipe height is taller.



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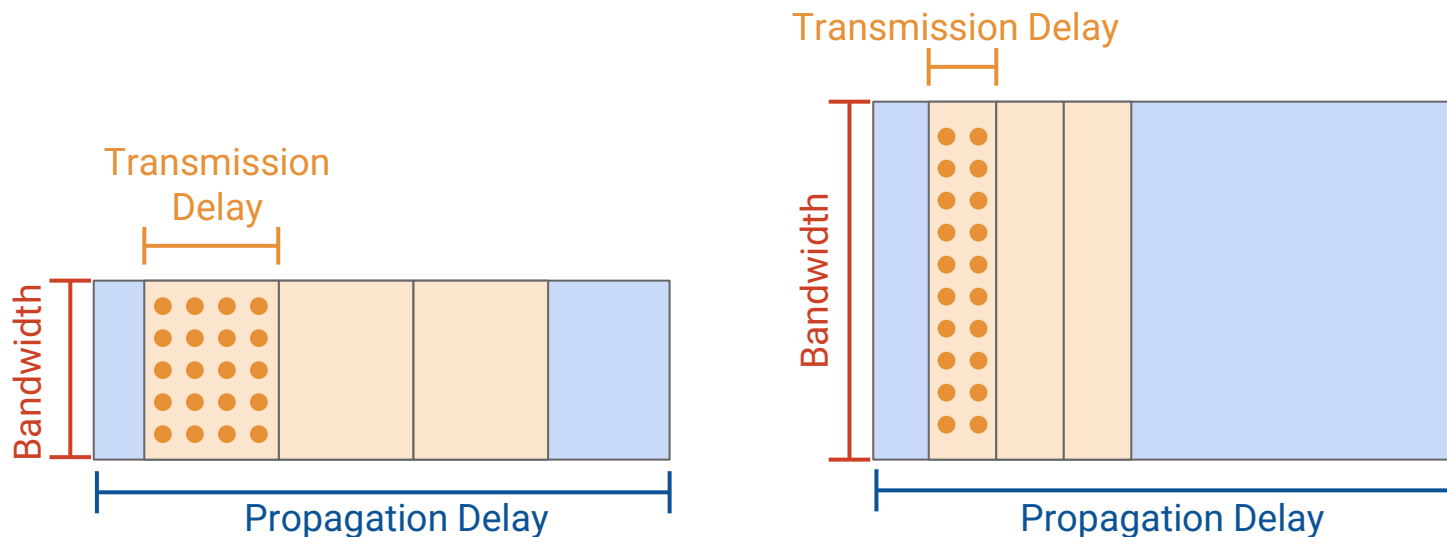
The pipe diagram shows the bits on the link at any given moment in time.

- **Height = bandwidth.** How many bits we can put in the pipe per unit time.
- **Width = propagation delay.** How long it takes for bits to travel through the pipe.
- **Area = bandwidth-delay product.** How many bits can fit in the pipe (bits in flight on the link) at a given instant.
- **Long & thin pipe:** satellite links → delay dominates, capacity small
- **Short & fat pipe:** LAN Ethernet → delay negligible, capacity huge
- **Long & fat pipe:** transoceanic fiber backbones → delay long, but capacity enormous
- **Short & thin pipe:** local low-rate links (serial bus, IoT, legacy modem) → delay tiny, but bandwidth the bottleneck



The **width of the packet** in the pipe represents the transmission delay.

- How long it takes to push all the bits onto the link.
- **Link capacity = Propagation Delay × Bandwidth**
- **Packet size = Transmission Delay × Bandwidth**
 - For a **packet of given size**: higher bandwidth → shorter transmission delay (higher bandwidth = taller pipe = more bits in pipe per unit time = narrower packet in pipe).



Overloaded Links

Lecture 3, Spring 2026

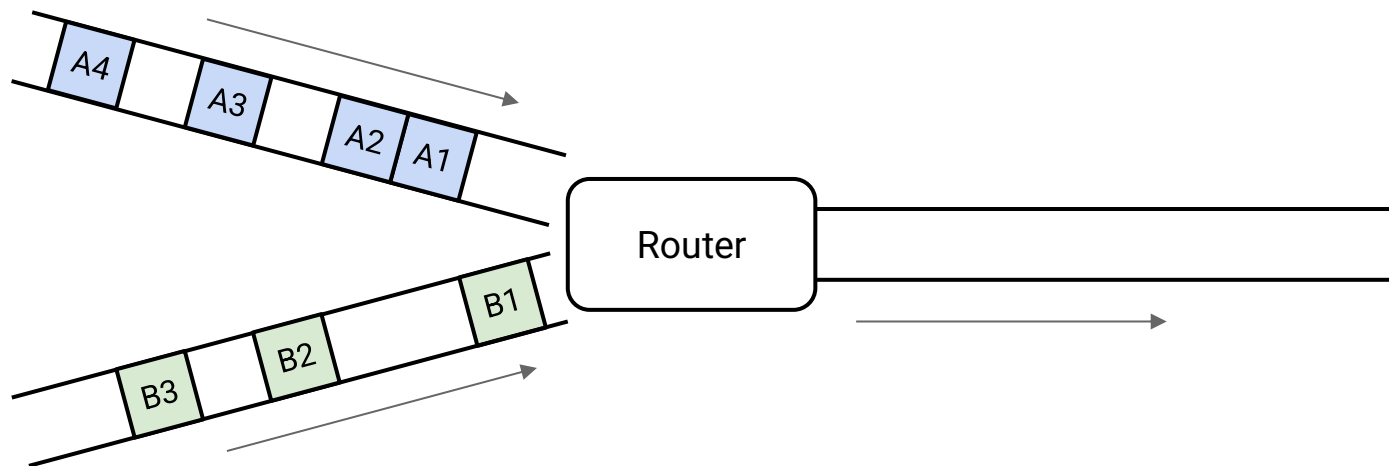
Links

- Bandwidth and Propagation Delay
- Pipe Diagrams
- **Overloaded Links**

Packet Switching at Routers

Recall: Routers receive packets, and forward them toward their destinations.

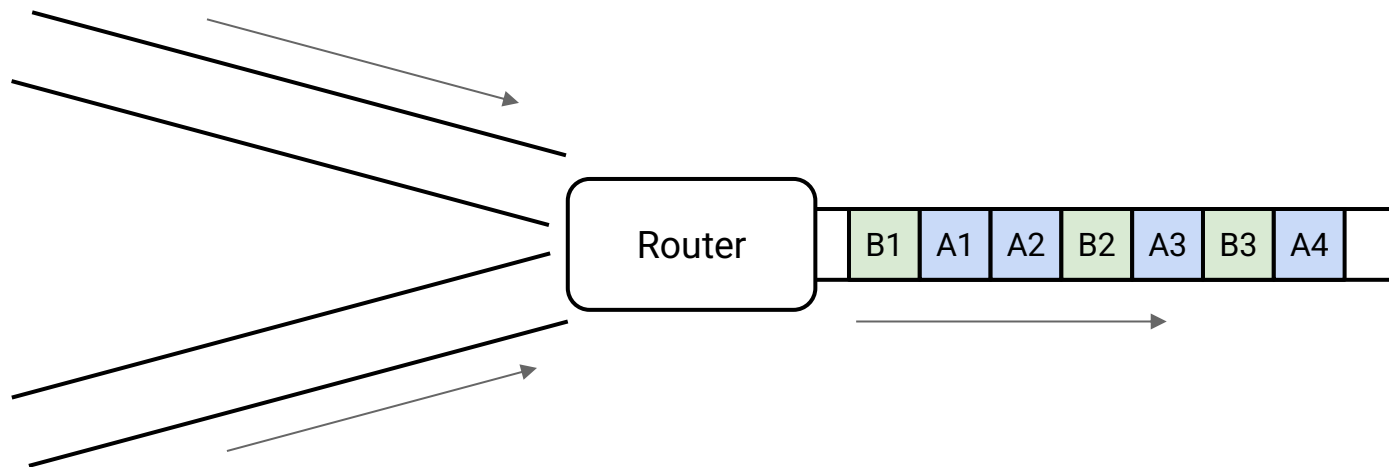
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- For simplicity, consider sending all outgoing traffic out of 1 link.



Packet Switching at Routers

Recall: Routers receive packets, and forward them toward their destinations.

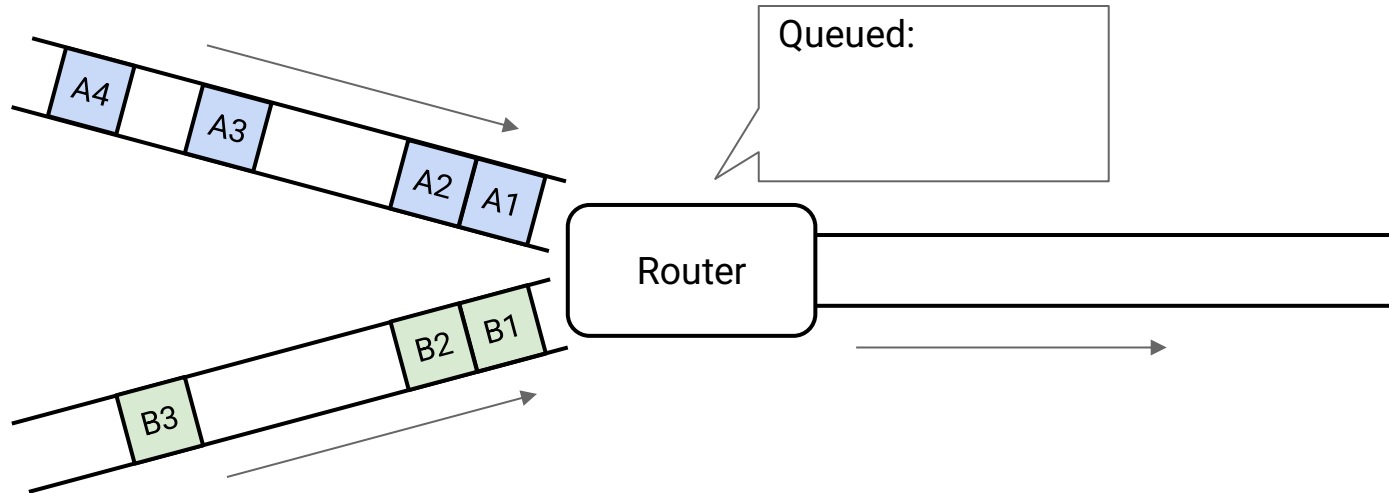
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Transient Overload

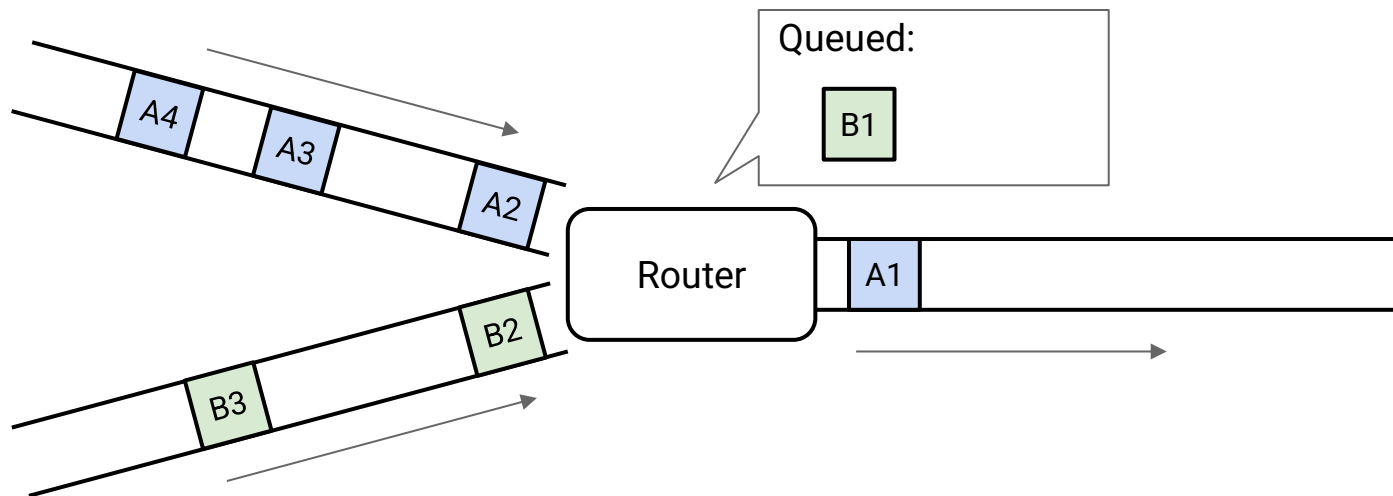
What happens if two packets arrive at the router simultaneously?

- Can't process both at the same time! Router must queue one for later.
- When there are no incoming packets, router can *drain* the queue.
- This is called **transient overload**, and it's fairly common.



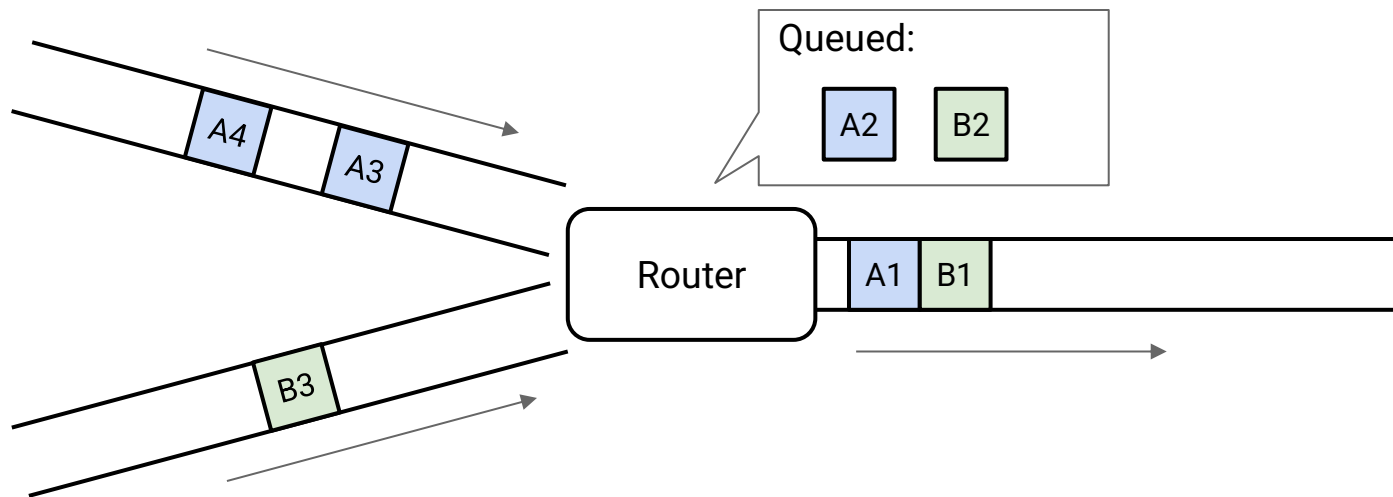
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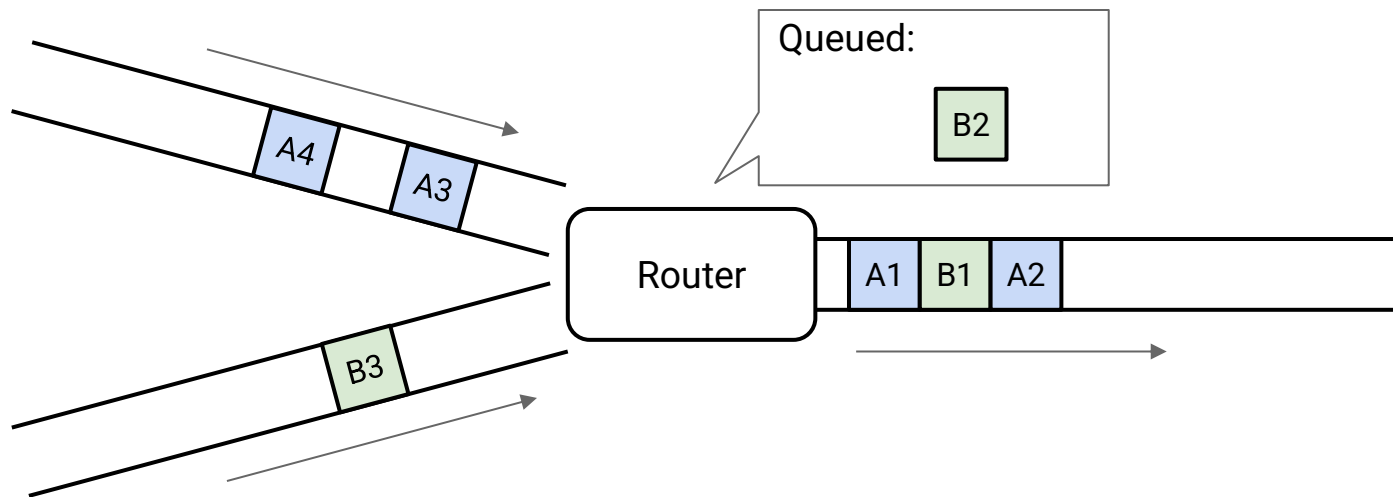
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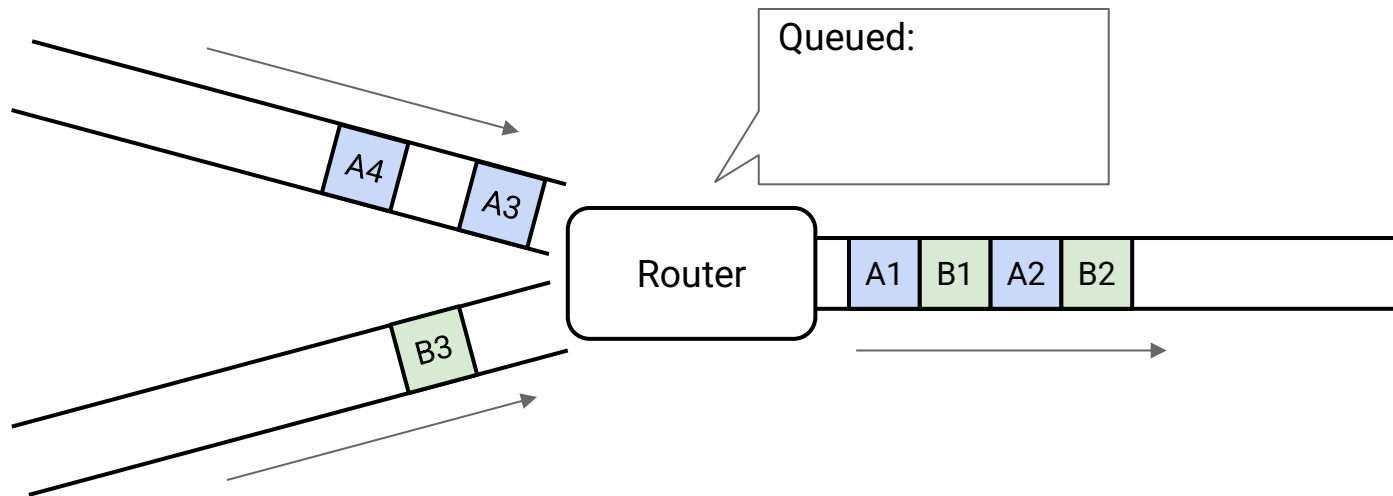
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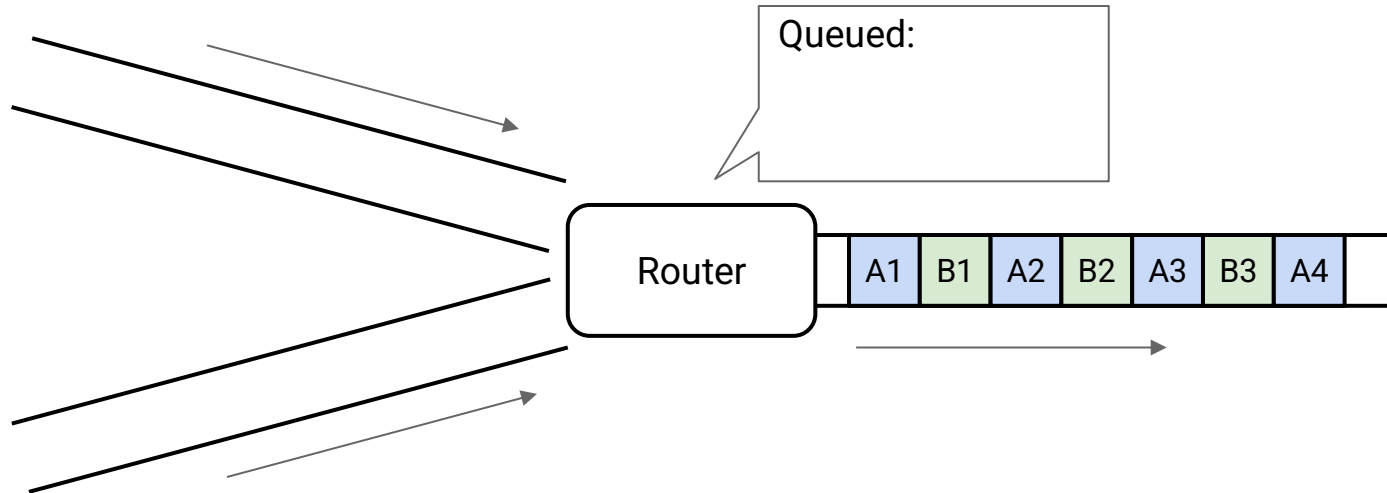
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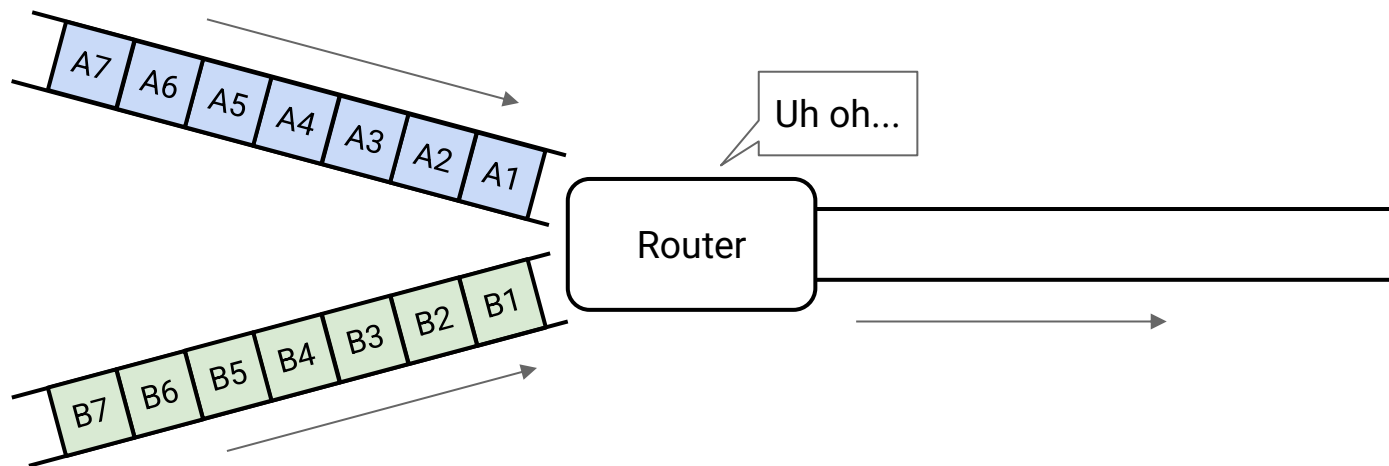


Persistent overload: Not enough capacity to handle the incoming packets!

- Queue won't help us. If the queue fills up, the router must drop packets.

How do we solve persistent overload?

- Operators can detect the overload and (manually) upgrade the link.
- Routers can tell the senders to slow down (throttling).



Life of a packet:

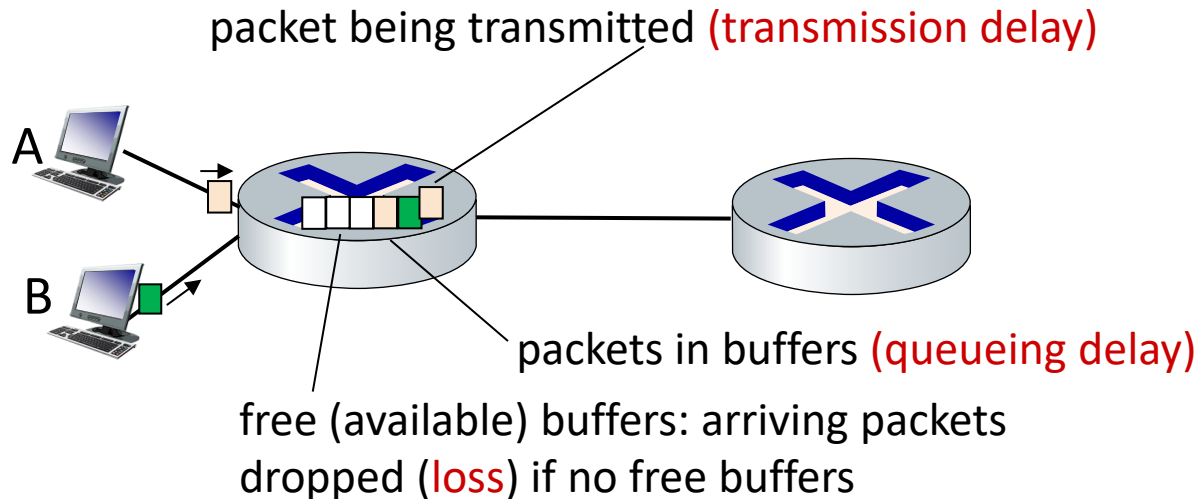
- Sender puts payload in a packet, adding headers.
- Packet travels along a link.
- Packet arrives at a router. Router forwards packet to the next hop.
 - **Store-and-forward**: A router can start forwarding a packet only after it has received all bytes of the packet.
 - Packet might be queued or dropped.
- Repeat the last step until:
 - Packet reaches destination.
 - Packet is dropped.

- How long does it take for your packet to travel through the network?
 - Packet delay = Transmission Delay + Propagation Delay + **Queuing Delay**
- It depends on...
 - **how much data** you're sending and the **link speed**
→ **transmission delay**
 - your **distance** from the destination
→ **propagation delay**
 - the **traffic pattern**
→ **queuing delay**

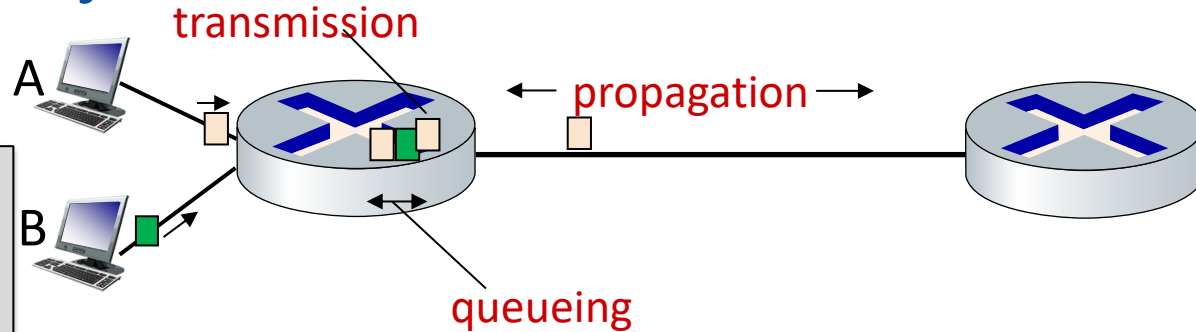
How do packet delay and loss occur?

Packets queue in router buffers, waiting for turn for transmission

- queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet **loss** occurs when memory to hold queued packets fills up



Packet delay: four sources



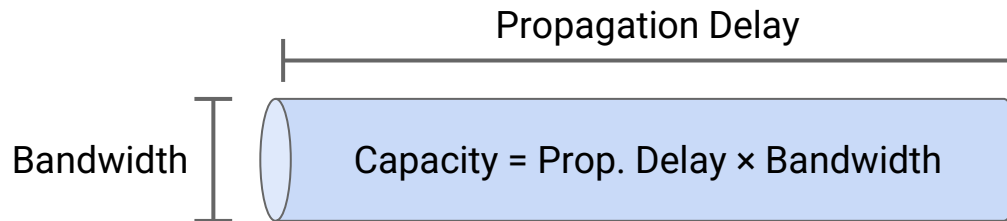
The nodal CPU processing delay (check bit errors, determine output link) is typically very small and can be ignored

$$d_{\text{nodal}} = d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

- | | | |
|---|---|---|
| <p>d_{queue}: queueing delay</p> <ul style="list-style-type: none"> time waiting at output link for transmission depends on congestion level of router | <p>d_{trans}: transmission delay:</p> <ul style="list-style-type: none"> L: packet length (bits) R: link <i>transmission rate</i> (bps) $d_{\text{trans}} = L/R$ | <p>d_{prop}: propagation delay:</p> <ul style="list-style-type: none"> d: length of physical link s: propagation speed ($\sim 2 \times 10^8$ m/sec) $d_{\text{prop}} = d/s$ |
|---|---|---|

Transmission Delay

- How long it takes for the all bits in the packet to get on the wire
 - The time between when the first and last bits enter the link
- Limited by the **bandwidth**
 - **Bandwidth**: Number of bits you can send through a wire per unit of time

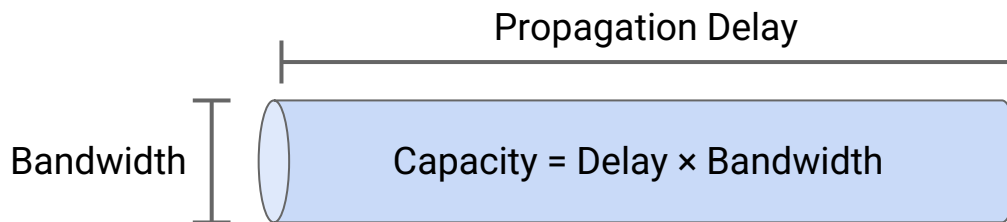


$$\text{Transmission Delay} = \frac{\text{packet size (bytes)}}{\text{bandwidth}}$$

bits/second

Propagation Delay (latency)

- End-to-end transmission time of one bit
- Depends on the **length of the link**
- Limited by propagation speed of link (speed of light)
- Does *NOT* depend on the size of the packet



$$\text{Propagation Delay} = \frac{\text{length of link (meters)}}{\text{speed of light (meters/second)}}$$

$$\text{e.g., Propagation Delay} = \frac{6000\text{km}}{3 \cdot 10^5 \frac{\text{km}}{\text{s}}} = 20\text{ms}$$

Summary: Links

- Packet Delay = $\underbrace{(\text{Packet Size} / \text{Bandwidth})}_{\text{Transmission Delay}} + \text{Propagation Delay} + \text{Queuing Delay}$
- Routers experience transient overload if packets arrive simultaneously.
Solution: Packets get queued for later.
- Routers experience persistent overload if there's insufficient capacity.
Queue gets full, and packets get dropped.

